

DATA COMMUNICATION AND COMPUTER NETWORKS

DCN Notes

Theory Notes

UNIT I — FUNDAMENTALS OF DATA COMMUNICATION AND COMPUTER NETWORK

1.1 Process of Data Communication and Its Components

Data communication is the exchange of digital or analog data between two or more devices through a communication medium. A communication system is designed so that information generated at one point can reach another point accurately and efficiently.

- **Sender (Transmitter):** The device that originates the data and prepares it for transmission.
- **Receiver:** The destination device that accepts the transmitted information.
- **Message:** The actual information being communicated, such as text, numbers, audio, images or video.
- **Transmission Medium:** The physical or wireless path through which the signal travels.
- **Protocol:** A set of rules that controls communication, including data format, timing, addressing and error handling.

A typical communication process is: source creates data → transmitter converts data into a suitable signal → medium carries the signal → receiver accepts and interprets it → destination obtains the information.

1.2 Protocols, Standards, Bandwidth, Data Transmission Rate, Baud Rate and Bits per Second

Protocol: A protocol defines how devices communicate. It specifies syntax (format), semantics (meaning) and synchronization (timing). Examples include TCP, IP, HTTP and Ethernet protocols.

Standard: A standard provides commonly accepted technical rules so that equipment from different manufacturers can work together. Standards may be developed by organizations such as ISO, IEEE, ITU-T and ANSI.

- **Bandwidth:** In a communication channel, bandwidth describes the range of frequencies that the channel can carry. In digital networking, the term is also commonly used informally for the capacity of a link.
- **Data Transmission Rate:** The amount of data transferred per second, generally expressed in bits per second (bps), Kbps, Mbps or Gbps.
- **Baud Rate:** The number of signal symbols transmitted per second. A symbol can represent one or more bits, so baud rate and bit rate are not always equal.
- **Bit:** The smallest unit of digital information, having a value of 0 or 1.

For a simple binary signalling system where one symbol represents one bit, bit rate and baud rate have the same numerical value. With multi-level signalling, one symbol can represent multiple bits, making the bit rate higher than the baud rate.

1.3 Modes of Communication

- **Simplex:** Communication occurs in only one direction. One device only sends and the other only receives. Example: traditional broadcast television.
- **Half Duplex:** Both devices can transmit, but not at the same time. Example: walkie-talkie communication.
- **Full Duplex:** Both devices can transmit simultaneously. Example: a telephone conversation.

Mode	Direction	Simultaneous?	Typical example
Simplex	One way	No	Broadcast system
Half Duplex	Both ways	No	Walkie-talkie

Mode	Direction	Simultaneous?	Typical example
Full Duplex	Both ways	Yes	Telephone / switched network

1.4 Analog Signal and Digital Signal; Analog-to-Digital and Digital-to-Analog Conversion

Analog Signal: A continuous signal that can take a continuous range of values. Speech from a microphone is naturally analog.

Digital Signal: A signal represented using discrete levels, commonly binary 0 and 1. Digital systems are easier to store, process and regenerate accurately.

Analog-to-Digital Conversion (ADC): Converts an analog signal into digital data. The main stages are sampling, quantization and encoding. Sampling measures the signal at regular intervals; quantization assigns each sample to a discrete level; encoding represents the quantized values as binary data.

Digital-to-Analog Conversion (DAC): Converts digital data into an analog signal. It is required when digital information must be transmitted through a medium or system that uses analog signalling.

1.5 Fundamentals of Computer Network

A computer network is a collection of interconnected devices that communicate and share resources using communication links and protocols. Networking allows users and systems to exchange information and share services.

- **Need:** Resource sharing, communication, centralized or distributed services, data sharing, remote access and improved collaboration.
- **Applications:** Web services, email, file sharing, online learning, cloud services, banking, multimedia communication and distributed applications.
- **Benefits:** Resource sharing, faster communication, scalability, centralized management, reliability and cost-effective sharing of hardware and software.

1.6 Classification of Network: LAN, MAN and WAN

Network	Coverage	Typical use	Main characteristic
LAN	Small area	Home, office, campus	High speed and locally managed
MAN	City / metropolitan area	Interconnecting LANs	Larger than LAN, smaller than WAN
WAN	Large geographic area	Country / worldwide networks	Uses long-distance communication links

UNIT II — TRANSMISSION MEDIA AND SWITCHING

2.1 Communication Media: Guided Transmission Media

Transmission media provide the path between communicating devices. Guided media use a physical conductor or cable. The major types are twisted-pair cable, coaxial cable and fibre-optic cable.

Twisted-Pair Cable

Twisted-pair cable consists of two insulated copper wires twisted together. Twisting reduces electromagnetic interference and crosstalk. It is widely used in telephone systems and Ethernet networks.

- **UTP (Unshielded Twisted Pair):** Lightweight and economical; commonly used in LAN installations.
- **STP (Shielded Twisted Pair):** Includes additional shielding to reduce electromagnetic interference.
- **Advantages:** Low cost, easy installation and flexible handling.
- **Limitations:** More susceptible to interference and attenuation than fibre for long-distance, high-capacity links.

Coaxial Cable

Coaxial cable has a central conductor, insulating layer, metallic shield and outer protective jacket. The shielding gives it better resistance to interference than ordinary twisted pair. It has been widely used in cable television and older network systems.

Fibre-Optic Cable

Fibre-optic cable transmits information as pulses of light through glass or plastic fibre. It consists of a core, cladding and protective coating. Light remains guided through the core because of the difference in refractive indices between the core and cladding.

- **Advantages:** Very high bandwidth, low attenuation, long-distance transmission, electrical isolation and strong resistance to electromagnetic interference.
- **Limitations:** Higher installation complexity, specialized splicing and testing equipment, and comparatively higher initial cost.

Medium	Signal	Interference resistance	Typical strengths
Twisted pair	Electrical	Moderate	Low cost, easy installation
Coaxial	Electrical	Better than ordinary twisted pair	Shielding and good signal quality
Optical fibre	Light	Very high	High bandwidth and long distance

2.2 Unguided Transmission Media

Unguided or wireless media transmit electromagnetic waves through air or space rather than a physical cable. Major examples are radio waves, microwaves, infrared and satellite communication.

- **Radio Waves:** Useful for wireless broadcasting and communication over relatively broad areas. They can propagate through air and, depending on frequency, may travel beyond direct line of sight.
- **Microwaves:** High-frequency electromagnetic waves used in point-to-point terrestrial links and other wireless systems. Many microwave links require a clear line of sight.
- **Infrared:** Short-range communication using infrared radiation. It generally does not pass through walls, making it useful for room-scale applications.
- **Satellite:** A communication satellite receives, amplifies or processes signals and retransmits them over a wide geographic area. It is useful for long-distance communication and broadcasting.

2.3 Line-of-Sight Transmission, Point-to-Point and Broadcast

Line-of-Sight (LOS): The transmitting and receiving antennas need an unobstructed path for effective communication. Buildings, terrain and the Earth's curvature can affect LOS links.

Point-to-Point: A dedicated communication path connects one transmitter with one receiver or one network endpoint with another.

Broadcast: A single transmission is intended for multiple receivers within the coverage area.

2.4 Multiplexing

Multiplexing allows multiple signals to share a common communication link. A multiplexer combines several input signals for transmission, while a demultiplexer separates them at the receiving side.

Frequency-Division Multiplexing (FDM)

In FDM, the available frequency spectrum is divided into separate frequency bands. Each signal uses a different band, and guard bands may be used to reduce interference between adjacent channels.

Time-Division Multiplexing (TDM)

In TDM, users share the same channel by transmitting during different time slots. In synchronous TDM, time slots are assigned in a fixed pattern; statistical TDM can allocate slots dynamically according to demand.

2.5 Switching

Switching determines how data moves through an intermediate network from source to destination. The syllabus includes circuit switching and packet switching.

Circuit-Switched Network

A dedicated logical path is established before communication begins. Resources along the path remain reserved for the duration of the session. It provides predictable communication once the circuit is established, but reserved capacity can remain unused during silent periods.

Packet-Switched Network

Data is divided into packets. Packets are forwarded through the network using addressing and routing information. Different packets may take different paths and share network resources with other traffic. Packet switching is efficient for bursty data communication and forms the basis of modern computer networks.

Circuit Switching	Packet Switching
Dedicated path is established	No dedicated end-to-end circuit is required
Resources are reserved	Resources are shared
Suitable for continuous traffic	Efficient for bursty data
Call/setup phase is required	Packets can be forwarded as they become available
Fixed path during the session	Packets may follow different routes

UNIT III — ERROR DETECTION AND CORRECTION

3.1 Types of Errors; Forward Error Correction versus Retransmission

An error occurs when received data differs from the data transmitted. Errors may be caused by noise, interference, attenuation or other transmission problems.

- **Single-Bit Error:** Only one bit in a data unit is changed.
- **Burst Error:** Two or more bits within a sequence are affected. Burst errors are common in practical communication systems.

Forward Error Correction (FEC): The sender adds redundant information so the receiver can detect and correct certain errors without asking for retransmission. It is useful where retransmission is costly or slow.

Retransmission / ARQ: The receiver detects an error and requests that the affected data be sent again. It generally requires feedback and can be efficient when errors are relatively infrequent.

3.2 Framing: Fixed-Sized and Variable-Sized Framing

Framing divides a continuous stream of bits into manageable units called frames. Frames provide boundaries so the receiver can identify the beginning and end of each transmitted unit.

Fixed-Sized Framing: Every frame has the same length. The receiver can identify frame boundaries using the known frame size.

Variable-Sized Framing: Frames can have different lengths. Delimiters, length fields, or other framing mechanisms are required to identify boundaries.

3.3 Error Detection: Repetition Code, Parity Bits, Checksums and CRC

Repetition Code: The same information is transmitted multiple times. The receiver can compare repetitions and select the value that appears most consistently. It is simple but adds substantial redundancy.

Parity Bit: One extra bit is added to make the number of 1s even (even parity) or odd (odd parity). A simple parity bit can detect many single-bit errors but cannot reliably detect all multiple-bit error patterns.

Checksum: Data is divided into fixed-size words and combined using arithmetic operations. The resulting checksum is transmitted with the data; the receiver recomputes it and compares the result.

Cyclic Redundancy Check (CRC): A bit sequence is treated as a polynomial and divided by a predefined generator polynomial using modulo-2 arithmetic. The remainder is appended to the frame. CRC is highly effective for detecting common transmission errors.

3.4 Error Correction: Automatic Repeat Request (ARQ) and Hamming Code

Automatic Repeat Request (ARQ): ARQ combines error detection with acknowledgements, timeouts and retransmissions. If a frame is damaged or lost, the sender transmits it again.

- **Stop-and-Wait ARQ:** One frame is sent and the sender waits for acknowledgement before sending the next frame.
- **Go-Back-N ARQ:** Multiple frames may be outstanding. If an error is detected, the sender retransmits the erroneous frame and subsequent frames in the affected sequence.
- **Selective Repeat ARQ:** Only missing or damaged frames are retransmitted, improving efficiency on links where errors are relatively uncommon.

Hamming Code: A block error-correcting code that inserts parity bits at selected positions. For a suitable number of parity bits, the receiver can locate and correct a single-bit error in a codeword.

3.5 Wireless LAN IEEE 802.11 Standard Architecture and Features

IEEE 802.11 is a family of standards for wireless local area networking. A basic WLAN can contain wireless stations and an access point. The access point can connect wireless stations to a distribution system or wired network.

- **802.11a:** Operates in the 5 GHz band and introduced higher data rates than the original standard.
- **802.11b:** Operates in the 2.4 GHz band and became widely used for early Wi-Fi networks.
- **802.11g:** Provides higher data rates in the 2.4 GHz band and supports compatibility with many 802.11b devices.
- **802.11n:** Introduced improvements such as MIMO and channel aggregation for higher throughput and better range.
- **802.11ac:** Focused on high-throughput operation in the 5 GHz band, using wider channels and advanced MIMO techniques.
- **802.11p:** Developed for wireless access in vehicular environments.

3.6 Bluetooth Architecture: Piconet and Scatternet

Bluetooth is a short-range wireless technology intended for personal-area communication. A **piconet** is a small Bluetooth network in which devices communicate under a common coordination arrangement. Multiple piconets can be interconnected to form a **scatternet**, allowing communication across more than one piconet.

3.7 Mobile Generations: 3G, 4G and 5G

Generation	General characteristics
3G	Mobile broadband, improved data services, multimedia communication and mobile Internet.
4G	Higher data rates, IP-based communication, lower latency and better support for multimedia and mobile broadband.
5G	Higher capacity, lower latency, support for massive device connectivity and advanced applications such as industrial IoT.

UNIT IV — NETWORK COMMUNICATION MODELS

4.1 OSI Model

The Open Systems Interconnection (OSI) model is a seven-layer reference model used to understand how network communication functions are organized. Each layer provides services to the layer above and uses services of the layer below.

Layer	Main functions
7. Application	Provides network services directly to user applications.
6. Presentation	Translation, data representation, compression and encryption-related functions.
5. Session	Establishes, manages and terminates communication sessions.
4. Transport	End-to-end delivery, reliability, flow control and segmentation.
3. Network	Logical addressing and routing of packets between networks.
2. Data Link	Framing, MAC addressing, error detection and local-link delivery.
1. Physical	Transmission of raw bits through the physical medium.

Encapsulation in the OSI Model

As data moves from the application layer toward the physical medium, each layer may add control information. This process is called encapsulation. At the receiver, the corresponding information is removed layer by layer, called decapsulation. The names commonly used for protocol data units are segment at the transport layer, packet at the network layer, frame at the data-link layer and bits at the physical layer.

4.2 TCP/IP Layers in Comparison with the OSI Model

The TCP/IP model is a practical networking model associated with the Internet protocol suite. A common four-layer representation contains Application, Transport, Internet and Network Access layers. Some references split the Network Access layer into Data Link and Physical layers, producing a five-layer teaching model.

TCP/IP layer	Typical responsibility	Related OSI layers
Application	Application-level network services and protocols	Application + Presentation + Session
Transport	Process-to-process delivery and reliability	Transport
Internet	Logical addressing and routing	Network
Network Access	Local delivery and physical transmission	Data Link + Physical

4.3 TCP/IP Protocols

- **HTTP:** Application-layer protocol used for web communication.
- **HTTPS:** HTTP carried over a secure cryptographic connection, commonly using TLS.
- **FTP:** Protocol designed for file transfer.
- **SMTP:** Used for sending and relaying email.
- **POP3:** Protocol for retrieving email from a mail server.
- **IMAP:** Protocol for accessing and managing email stored on a server.
- **DNS:** Resolves domain names to IP addresses and other DNS records.

- **DHCP:** Automatically provides hosts with configuration information such as IP address, subnet mask, gateway and DNS server.
- **ARP:** In IPv4 local networks, resolves an IP address to a link-layer address such as a MAC address.
- **ICMP:** Carries network-layer control and diagnostic messages.
- **TCP:** Connection-oriented transport protocol providing reliable, ordered byte-stream delivery.
- **UDP:** Connectionless transport protocol with low overhead and no built-in guarantee of delivery or ordering.

4.4 Addressing: Physical, Logical and Port Address

- **Physical Address:** A link-layer address used for delivery within a local network segment. Ethernet commonly uses MAC addresses.
- **Logical Address:** A network-layer address used for communication across interconnected networks. IPv4 and IPv6 provide logical addressing.
- **Port Address:** A transport-layer identifier used to distinguish application or service endpoints on a host. TCP and UDP use port numbers.

4.5 IPv4 Addressing: Notation, Address Space and Classes

An IPv4 address is 32 bits long and is commonly written in dotted-decimal notation, such as four decimal octets separated by dots. The address space contains 2^{32} possible bit patterns. Historically, classful addressing divided IPv4 addresses into Classes A, B, C, D and E. Modern networks generally use CIDR rather than relying on these fixed classes.

Class	Historical first-octet range	Traditional purpose
A	1–126	Very large networks
B	128–191	Medium-sized networks
C	192–223	Smaller networks
D	224–239	Multicast
E	240–255	Reserved / experimental

4.6 IP Address Concept, Notation and Address Space

An IP address identifies a network interface at the Internet Protocol layer. IPv4 uses 32 bits, while IPv6 uses 128 bits. IPv6 provides a vastly larger address space and was designed to address the limitations of IPv4 address availability, while also introducing other protocol improvements.

4.7 IPv4 Addressing: Classful and Classless Addressing, Subnet Mask, Supernetting and Subnetting

Classful Addressing: Uses fixed network/host boundaries based on address classes. It can waste address space because network sizes are restricted to class boundaries.

Classless Addressing (CIDR): Uses a prefix length such as /24 to specify how many leading bits identify the network. CIDR allows flexible allocation and route aggregation.

Subnet Mask: A 32-bit value used with IPv4 to identify the network and host portions of an address.

Subnetting: Divides one network into smaller logical networks by borrowing host bits for subnet identification.

Supernetting: Combines multiple contiguous networks into a larger address block, reducing the number of routing entries through route aggregation.

4.8 IPv6 Addressing Scheme and Basic Structure

IPv6 uses 128-bit addresses written as eight groups of hexadecimal digits separated by colons. Leading zeros within a group can be omitted, and one continuous sequence of all-zero groups can be compressed using :: once in an address. IPv6 supports unicast, multicast and anycast addressing and does not use IPv4-style broadcast addressing.

UNIT V — NETWORK TOPOLOGIES AND NETWORK DEVICES

5.1 Network Computing Model: Peer-to-Peer and Client-Server

Peer-to-Peer (P2P): Devices can act as both clients and servers and share resources directly. P2P is simple for small groups but can become difficult to manage and secure as the network grows.

Client-Server: Clients request services from dedicated servers. Servers provide centralized resources such as files, authentication, databases or web applications. This model offers centralized administration and is suitable for larger managed networks.

Parameter	Peer-to-Peer	Client-Server
Architecture	Decentralized	Centralized around servers
Management	Distributed	Centralized
Scalability	Limited for large organizations	Better suited to large networks
Cost	Generally lower for small networks	Higher due to server infrastructure
Security / control	More difficult to centralize	Centralized policies are easier to apply

5.2 Network Topologies

Network topology describes the arrangement of network devices and communication links. A good topology is selected according to cost, scalability, fault tolerance, performance, installation requirements and the physical environment.

Star Topology

All nodes connect to a central device such as a switch. A failure of one endpoint link generally affects only that endpoint, while failure of the central device can affect the whole network. Star topology is widely used in Ethernet LANs.

Mesh Topology

Nodes have multiple interconnections. A full mesh connects every node to every other node, while a partial mesh provides selected redundant links. Mesh offers strong fault tolerance but requires more cabling, interfaces and configuration.

Tree Topology

Tree topology organizes networks in a hierarchical structure. It can be viewed as a collection of star networks connected through higher-level devices. It supports expansion but depends on higher-level links and devices.

Hybrid Topology

Hybrid topology combines two or more topology types, such as star-bus or star-ring arrangements. It can be designed to match different parts of an organization but may increase design and maintenance complexity.

Topology	Main advantage	Main limitation
Star	Easy management and expansion	Central device is critical
Mesh	High redundancy and fault tolerance	High cost and complexity
Tree	Hierarchical expansion	Upper-level failure can affect branches
Hybrid	Flexible design	More complex planning and maintenance

5.3 Network Connecting Devices

Network connecting devices operate at different layers and perform different forwarding or signal-related functions.

Switch

A switch connects devices within a LAN and forwards Ethernet frames using MAC addresses. A switch learns which MAC addresses are reachable through its ports and uses this information to forward frames efficiently. Modern switches can also support VLANs, trunking, link aggregation and other network functions.

Router

A router connects different IP networks and forwards packets using logical addresses and routing information. Routers maintain routing tables and select appropriate next hops. They can also provide functions such as inter-network communication, filtering and address translation depending on the device and configuration.

Repeater

A repeater receives a signal and regenerates or retransmits it to extend the effective communication distance. It does not make routing decisions or understand application-level data.

Hub

A hub is a simple multiport device that repeats incoming signals to other ports. It does not selectively forward frames based on destination MAC addresses. Hubs are largely obsolete in modern Ethernet networks because switches provide better efficiency.

Device	Primary function	Typical layer / concept
Repeater	Regenerates and extends a signal	Physical layer
Hub	Repeats traffic to multiple ports	Physical layer
Switch	Forwards frames using MAC addresses	Data Link layer
Router	Forwards packets between IP networks	Network layer

QUICK REVISION REFERENCE

Key terms from the five units

- **Data Communication:** Exchange of information between communicating devices.
- **Protocol:** Rules governing communication.
- **Simplex / Half Duplex / Full Duplex:** One-way / two-way non-simultaneous / two-way simultaneous communication.
- **LAN / MAN / WAN:** Small-area / metropolitan / large-area networks.
- **Guided Media:** Twisted pair, coaxial and fibre optic.
- **Unguided Media:** Radio, microwave, infrared and satellite.
- **FDM / TDM:** Sharing a link by frequency bands / time slots.
- **Circuit Switching / Packet Switching:** Reserved path / shared packet-based forwarding.
- **Framing:** Dividing a bit stream into frames.
- **CRC:** Polynomial-based error-detection method.
- **ARQ:** Error recovery using acknowledgement and retransmission.
- **OSI:** Seven-layer reference model.
- **TCP/IP:** Practical protocol architecture used by the Internet.
- **IPv4 / IPv6:** 32-bit / 128-bit Internet addressing.
- **Topology:** Arrangement of network nodes and links.
- **Switch / Router / Repeater / Hub:** LAN forwarding / inter-network routing / signal regeneration / signal repetition.